S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Vitamin B_2** **Label Claim:** **1500mcg/5ml**	**METHOD 1:** Reagents: a. 0.1N Sodium Hydroxide b. Phosphate Buffer (1 M), pH 6.0, c. Glacial Acetic Acid d. 2% w/v solution of Potassium Permanganate e. Hydrogen Peroxide (30%) ***Standard solution**:* 150mcg/mL of Riboflavin phosphate CRS in 0.1N HCl ***Sample solution:*** Accurately weigh a quantity of the sample equivalent to 15mg of the compound and extract with acid to make 100mL Procedure**:** To 3ml each of the sample and standard solutions, add 1 drop of Phenolphthalien and neutralize with 0.1N sodium hydroxide. Add 10mL of buffer solution followed by 2mL of Acetic Acid, 1mL of Potassium Permanganate solution. Allow to stand for

		2 minutes, neutralize pink colour with 1mL of Hydrogen Peroxide. Shake vigorously and dilute to 25mL with water. Measure the Absorption of both the solutions at about 445nm and calculate the results by comparison **% Assay of Vitamin B_2** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 **Note:** The method is application to most of the Vitamin combinations and gives good results. ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
3.	**Vitamin B_1** **Label Claim:** **1500mcg/5ml**	**METHOD 1:** Reagents: a. 0.04%w/v solution of p-amoniphenol in ethyl alcohol. b. 0.1M Ammonium Hydroxide solution ***Standard solution:*** Dissolve 100mg of Thiamine Hydrochloride (previously dried at 100oC) in distilled water to produce 100mL. Dilute this solution suitably with water to give a

concentration of 20mcg/mL

Sample solution:

Dissolve and dilute suitably with water to give a concentration of 20mcg/mL

Procedure:

To 3ml each of the sample and standard solutions, add 3ml of 0.1N Ammonium Hydroxide solution and 0.5ml of p-aminophenol reagent. Wait for 5 minutes and extract the coloured complex with 10ml of Chloroform. Measure the Absorption of Organic layer at about 430nm against reagent blank. The contents of thiamine are calculated by comparison. The Method obeys Beer's law in the concentration range of 3 – 12 mcg/ml in final solution

The following compounds do not interfere in the above method. The figures in brackets indicate the number of folds excess the compounds can be tolerated.

- Pyridoxine
- Niacinamide
- Riboflavin
- Vitamin B12
- Ascorbic Acid

% Assay of Vitamin B_1 = $\frac{absorbance of sample}{absorbance of standard}$ x 100

		REFERENCE: Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
4.	**Ascorbic acid** **Label Claim:** **40mg/5ml**	**METHOD 1** **Procedure:** Weigh accurately, syrup equivalent to 150mg of ascorbic acid. Add to this, 10mL acetic acid and 20mL water. Titrate with 0.1N iodine solution using starch as indicator. Each mL of 0.1N iodine solution is equivalent to 0.008806g of ascorbic acid. $$\%\ \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor \quad x \quad 8.806}{150}\ x\ 100$$ **Note***: Iodine is a mild oxidizing agent. It oxidizes ascorbic acid to dehydroascorbic acid and endpoint being determined by the formation of blue-coloured iodine-starch inclusion complex.* **METHOD 2:** ***Standard*** Weigh accurately 150mg of Ascorbic Acid and dissolve in 0.01M HCl and make up to mark using a 100ml volumetric flask. Take exactly 1ml of the resulting solution and make up to mark of a 100ml volumetric flask using the same solvent. ***Sample***

	Note: 55mg due to overage	Take quantitatively equivalent of 150mg (exactly 5ml) of the Syrup and dissolve with 0.01M HCl and make up to mark using a 100ml volumetric flask. Take accurately 1ml of the resulting solution and make up to mark of a 100ml volumetric flask using the same solvent. **% Assay of Ascorbic Acid** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** *Sethi's* Quantitative Analysis of Pharmaceutical Formulations. Vol. 3 **METHOD 3:** **Procedure:** Weigh accurately, syrup equivalent to 40mg of ascorbic acid, add 10mL of dilute sulphuric acid and then 80mL of carbon dioxide-free water. Add 1mL of starch indicator. Titrate with 0.05M iodine until a persistent violet-blue colour is obtained. 1mL of 0.05M iodine is equivalent to 8.81mg of $C_6H_8O_6$. $$\% \, \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor \quad x \quad 8.81}{55}\ x\ 100$$ **REFERENCE:** *British Pharmacopoeia, 2024, Ascorbic acid: Ph. Eur. Monograph 0253.*
5.	**Nicotinamide Label Claim:**	**METHOD 1:** **Reagents:**

	10mg/5ml	a. Sulphanilic acid buffered solution: To 2.5g sulphanilic acid, add 15ml water and 2ml dilute ammonia solution. Mix, add with stirring more dilute ammonia solution if necessary to dissolve the substance, adjust pH of the solution to 4.5 with dilute ammonia b. 10% w/v solution of cyanogens bromide in water. c. Dilute ammonia solution: Dilute 2ml strong ammonia solution to 100ml with water. ***Standard solution:*** 100mcg/ml of Nicotinamide in water. ***Sample solution:*** Weigh equivalent to 10mg of the substance, suspend in water to make 100ml, stir and filter. **Procedure:** To 2ml each of sample and standard solutions, add 2ml of sulphanilic acid buffered solution, 5ml water, shake well and add 2ml of cyanogens bromide solution. Measure the entinction of both sample and standard solutions at 450nm after 2min against reagent blank and calculate the contents of Nicotinamide by comparison. **% Assay of Nicotinamide** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

6.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an “S” or “8” shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days. **<u>TEST FOR ABSENCE OF PATHOGEN</u>** - Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth - Transfer 10ml into 90ml of Tryptone soya broth - Incubate at 30-35^{0}c for 18-24hours

- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

		CFU/mL =<u>Total number of colonies obtained x dilution factor</u> **Volume of sample used (aliquot)** - This will give the total number of viable microbial cells present in the given sample. - For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated. - If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats. - The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

*LAF--- Laminar Air Flow

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BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR BRADVITE-M SYRUP (100ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/017/F01**	**PAGE NO.:**	**Page 10 of 10**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		